The Nesting Domain Handoff Problem

Parity Networks and the Modular Structure of Gravitational Scenes

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Abstract

General relativity gives an extraordinarily successful local description of gravitation through metric structure, curvature, and geodesic behavior. Yet every practical gravitational computation begins before the metric is evaluated: a scene must be declared. A scene includes a source roster, coordinate frame, time slice, approximation regime, readable-field definition, datum convention, causal access condition, and enclosing context. This paper argues that this act of declaration is not passive bookkeeping. It contains recoverable mathematical structure.

We introduce the *nesting domain handoff problem*: the problem of formalizing how a high-resolution child scene is embedded within, supported by, and interpreted through a lower-resolution parent context. Multiscale gravitational systems are not isolated. A moon belongs to a planet system, a planet system belongs to a stellar system, and a stellar system belongs to a galactic environment. The local equations may remain valid at each scale, while the operational grammar of parenthood, child identity, contextual support, boundary inheritance, and scale transition remains underdeveloped.

We propose that *parity networks*, derived from equality relations between declared gravitational contributions and their derived context at a declared time slice, expose a modular organization of gravitational scenes. This modularity is not a discontinuity in the gravitational field. It is a relational decomposition of the declared scene: a way to identify source-context equality, child domains, parent supports, and candidate handoff regions.

The central claim is that parity-network hierarchies provide the seed of a *context calculus* for multiscale gravitational systems. We introduce formal definitions for gravitational scenes, declared contributions, derived contexts, parent-context uncertainty, handoff confidence, tidal orientation witnesses, and handoff functionals. We predict that parity-network hierarchy will correlate with transitions in tidal eigenstructure, continuity-preserving released-lattice deformation, and trajectory diagnostics in controlled weak-field systems.

The local equations may be correct while inherited bookkeeping of gravitational context remains incomplete.

1 Introduction

General relativity supplies one of the most profound descriptions of nature ever developed. It gives the mathematical machinery of metric structure, curvature, geodesic motion, and local gravitational dynamics. Its success is not in question here.

This paper begins elsewhere.

Before a gravitational field is computed, a scene must be declared. One must decide which sources are included, which are omitted, which are treated as parent context, which time slice is being analyzed, which coordinates are being used, what approximation regime applies, what can be causally known, and which subsystem is being resolved.

These choices are usually treated as pragmatic modeling decisions. They are necessary, but not usually elevated into the mathematical structure of the analysis itself.

The claim of this paper is that this is a missed opportunity.

A declared gravitational scene has internal relational structure. It is not merely a list of sources inside a coordinate box. It contains contribution-context relations, child-parent relations, parity boundaries, handoff supports, uncertainty gradients, and modular organization. These structures do not modify the equations of general relativity. They describe how gravitational context is organized before, during, and after those equations are applied.

This paper proposes that parity networks make this hidden structure visible.

The present work extends a four-part Gravitational Coherence Surface sequence. Paper I introduced the GCS as a parity horizon of gravitational distinguishability under a declared readability rule; Paper II generalized the shell logic into many-source shared parity networks; Paper III argued that a mesoscale readability layer can persist between pointwise field evaluation and coarse averaging; and Paper IV reported proof-of-concept extraction of many-source boundary-support geometry in standard weak-field settings [1, 2, 3, 4].

A parity network is a derived structure built from equality relations between gravitational contributions and their declared context. At a time slice, it can reveal how a scene decomposes into contextual modules: sources, groups, parent supports, child domains, and handoff regions.

The proposal is not that the gravitational field is discontinuous. The proposal is that *scene interpretation has modular structure.*

That modularity has consequences.

If a local child scene is computed with high precision while its parent context is treated coarsely, the relation between them becomes mathematically important. If averaging erases the boundary at which child identity hands off to parent context, it may erase structure that matters for tidal organization and trajectory behavior. If a parity-network hierarchy remains stable under controlled perturbation, it may provide a new diagnostic layer for multiscale gravitational modeling.

The guiding thesis is simple:

> General relativity supplies the local law. A context calculus is needed to formalize how declared gravitational scenes nest, hand off, and preserve identity across scale.

This paper does not propose a new force law. It does not propose a replacement for the Einstein field equations. It proposes something more modest in mechanism but potentially broad in consequence: a mathematical grammar for the relational structure of gravitational scenes.

If such a grammar is stable, it would not compete with GR's local machinery. It would clarify how that machinery is applied to multiscale systems whose relevant structure is lost when context is declared too coarsely.

2 Gravitational Scenes

A gravitational scene is not the universe. It is a declared computational and interpretive domain.

We define a gravitational scene as

$$S_O(t) = (\Omega, \Sigma, D, R, F, C_O, \Pi, \Lambda), \tag{1}$$

where Ω is the declared coordinate domain, Σ is the source roster, D is the datum or registration convention, R is the set of readable-field definitions, F is the field, metric, or diagnostic operator set, C_O is the observer's causal access condition, Π is the parent-context declaration, and Λ is the confidence record for the scene.

This definition elevates scene declaration into a formal object.

A scene is not merely where the computation happens. It defines what the computation means.

The observer enters through C_O, not as an invitation to subjectivism, but as an operational constraint. Observers do not receive a complete universe at infinite resolution. They receive signals, construct models, declare domains, and compute within causal limits.

In the present weak-field, single-time-slice formulation, C_O is a declared constraint rather than a dynamically propagated light-cone operator. Retardation, light-cone reconstruction, and finite-signal parent updates are reserved for the relativistic extension of the framework. This keeps the present paper focused on scene declaration and handoff diagnostics rather than causal reconstruction.

Thus, a gravitational scene is partly objective and partly declared.

The source masses, separations, field values, metric terms, and tidal tensors are not arbitrary. But the choice of scene, parent context, resolution, and causal access condition is part of the formal structure of the computation.

A scene is therefore a disciplined claim of relevance.

3 Nested Scenes and the Handoff Problem

Gravitational systems are nested.

A moon belongs to a planet system. A planet belongs to a stellar system. A stellar system belongs to a local stellar neighborhood. A stellar neighborhood belongs to a galaxy. Each domain may obey the same local gravitational law, but each domain changes the context in which local structure is interpreted.

This creates the *nesting domain handoff problem.*

A child scene may be internally high-confidence. Its source roster may be well known. Its local distances may be precise. Its internal field structure may be sampled densely.

Yet the child scene is embedded within a parent context that is larger, coarser, causally delayed, incompletely rostered, or modeled at lower resolution.

So the confidence of the child and the confidence of the parent are not generally equal:

$$L_{\text{child}} \neq L_{\text{parent}}. \tag{2}$$

More importantly, the confidence of the handoff between them is not identical to either one:

$$L_{\text{handoff}} \neq L_{\text{child}}, \qquad L_{\text{handoff}} \neq L_{\text{parent}}. \tag{3}$$

The child can be known sharply. The parent can be known broadly. The relation between them must be evaluated.

We propose a first handoff-confidence relation:

$$L_{\text{handoff}} = L_{\text{child}} \cdot L_{\text{parent}} \cdot R_{\text{stability}} \cdot R_{\text{boundary}}, \tag{4}$$

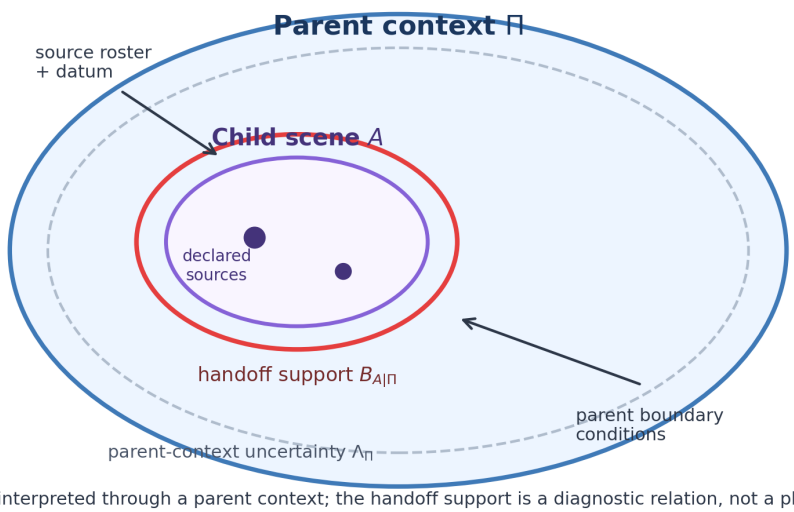
where $R_{\text{stability}}$ measures persistence under controlled perturbations, and R_{boundary} measures the coherence of the child-parent boundary or support relation.

This is the first major formal claim:

> A nested gravitational scene requires a confidence-weighted handoff relation between child and parent domains.

Without that relation, multiscale modeling risks treating the child as isolated or the parent as a smooth background too early.

Both choices can erase structure.



A nested scene is interpreted through a parent context; the handoff support is a diagnostic relation, not a physical membrane.

Figure 1: Conceptual schematic of a nested gravitational scene. A child scene A is interpreted relative to an enclosing parent context Π. The handoff support $B_{A|\Pi}$ is not a physical membrane; it is a derived diagnostic region where child-parent interpretation, boundary inheritance, and confidence accounting become explicit.

4 Parent-Context Uncertainty

A parent context may be necessary for interpreting a child scene while remaining unresolved at the child's resolution.

We define *parent-context uncertainty* as the portion of enclosing gravitational structure required for interpreting a child scene but not available at the same resolution, completeness, or causal freshness as the child scene itself.

Let the parent context be decomposed into known and unresolved portions:

$$\Pi = \Pi_k \cup \Pi_u, \tag{5}$$

where Π_k is the known or declared parent context, and Π_u is the unresolved parent contribution.

A first contribution-based uncertainty measure is

$$\Lambda_\Pi(x) = \frac{Q^{\max}_{\Pi_u}(x)}{Q_{\Pi_k}(x) + \epsilon}. \tag{6}$$

The numerator in Eq. (6) should be read as a prior-bound or completeness-bound estimate, not as a directly computed contribution from a known roster. If sources in Π_u were fully known, they would belong in Π_k. In practice, $Q^{\max}_{\Pi_u}$ must be derived from declared assumptions such as mass cutoffs, survey completeness limits, maximum unresolved background contribution, or other scene-specific priors. This creates an unavoidable modeling loop: estimating unresolved parent contribution requires assumptions about the very context whose incompleteness the term is meant to mark. The value of Λ_Π is therefore not an oracle. It is an explicit confidence penalty.

More generally, $\Lambda_\Pi(x)$ represents parent-context uncertainty at x.

Parent confidence is then reduced by unresolved context:

$$L_\Pi(x) \propto \frac{1}{1 + \Lambda_\Pi(x)}. \tag{7}$$

This gives a formal way to state a common modeling fact: a child scene may be internally precise while its enclosing parent scene remains partially uncertain.

The parent is not irrelevant. It is simply not known with equal sharpness.

That asymmetry is not a nuisance. It is a defining feature of nested gravitational computation.

5 Declared Contributions and Derived Contexts

The proposed framework begins with a readable contribution field. In mass-normalized form:

$$Q_i(x) = \frac{M_i}{r_i(x)^3}, \tag{8}$$

where M_i is the mass of source i, and $r_i(x)$ is the distance from source i to the query point x, with cutoff or softening rules explicitly declared.

A dimensional tidal-scale form is

$$\mathcal{Q}_i(x) = \frac{GM_i}{r_i(x)^3}. \tag{9}$$

The distinction is simple. Q_i is a readable contribution diagnostic. $\mathcal{Q}_i$ carries the dimensional gravitational scale associated with tidal falloff. Both are derived diagnostic quantities, not new force laws.

For source i, define the derived context as the contribution of all other declared sources:

$$Q_{\neg i}(x) = \sum_{j \neq i} Q_j(x). \tag{10}$$

The source-context residual is

$$\Psi_i(x) = Q_i(x) - Q_{\neg i}(x). \tag{11}$$

The corresponding parity boundary is

$$G_i = \{x \in \Omega : \Psi_i(x) = 0\}. \tag{12}$$

This is the surface where a declared contribution reaches equality with its derived context.

For pairwise parity:

$$\Psi_{ij}(x) = Q_i(x) - Q_j(x), \tag{13}$$

and

$$G_{ij} = \{x \in \Omega : \Psi_{ij}(x) = 0\}. \tag{14}$$

A family of such relations forms a parity network.

The key idea is that context is not assigned arbitrarily. It is derived from the declared scene.

6 Parity Networks

At a declared time slice, a parity network is a graph or family of surfaces derived from equality relations between declared gravitational contributions and derived contexts.

The central claim is:

> Parity networks reveal a modular organization of gravitational context within continuous gravitational scenes.

This does not mean the field itself is broken into pieces. It means the declared scene admits relational decomposition.

For a child group A and parent context Π:

$$Q_A(x) = \sum_{i \in A} Q_i(x), \tag{15}$$

$$Q_\Pi(x) = \sum_{j \in \Pi} Q_j(x), \tag{16}$$

and

$$\Psi_{A|\Pi}(x) = Q_A(x) - Q_\Pi(x). \tag{17}$$

The group-parent parity boundary is

$$G_{A|\Pi} = \{x : \Psi_{A|\Pi}(x) = 0\}. \tag{18}$$

This boundary is not a physical membrane. It is a *parity handoff surface*: a derived surface where the declared child contribution reaches equality with its parent context.

We reserve the term *functional handoff support* for regions later identified by a multi-witness handoff functional. In this notation, $G_{A|\Pi}$ is the parity handoff surface, while $B_{A|\Pi}$ is the functional handoff support.

In this sense, a parity network is a modular scene-decomposition diagnostic.

It can identify child domains, parent supports, source-context equality regions, group-context transitions, candidate handoff boundaries, and regions where averaging may erase structure.

A parity network does not replace the metric. It reveals relational organization within the declared scene to which the metric is applied.

7 Relation to Existing Concepts

The proposed framework is adjacent to several established gravitational concepts. Its novelty does not come from noticing that gravitational systems are nested. Its novelty lies in treating *declared scene modularity* and *child-parent handoff* as first-class diagnostic objects.

Hill spheres and gravitational spheres of influence identify regions where a smaller body's gravitational influence dominates over perturbations from a larger body in an orbital setting. They are powerful and physically meaningful, especially in restricted three-body contexts. However, a Hill sphere is not a general scene-decomposition formalism. It does not by itself encode arbitrary source rosters, parent-context uncertainty, parity-network hierarchy, or confidence-weighted handoff between nested domains [5].

Roche lobes and Roche limits describe tidal disruption, mass transfer, and gravitational binding constraints in specific two-body or orbital systems. These are important boundary concepts, but they are not designed to extract modular source-context structure from general multiscale scenes.

Lagrange points identify equilibrium points in rotating frames, especially in restricted three-body systems. They reveal special dynamical structure, but they do not provide a general grammar for source identity, parent support, or child-domain inheritance across arbitrary declared scenes.

Multipole expansions compress source structure into moments and are indispensable for far-field approximation. But compression can hide the very modular structure this framework seeks to preserve. A multipole representation may preserve large-scale field behavior while erasing child-domain identity, parity hierarchy, and handoff structure.

Tidal tensors and geodesic deviation describe local relative deformation. They are central to this proposal. However, they do not by themselves define which subsystem is a child scene, which enclosing structure is a parent context, or where interpretation changes scale. In this paper, tidal eigenstructure becomes an orientation witness inside a broader handoff diagnostic.

Averaging and coarse-graining methods are necessary in multiscale modeling [6, 8]. But averaging often begins after one has already decided what can be safely compressed. The present framework asks a prior question: which relational structures should be retained before averaging proceeds? This is especially relevant because averaging and evolution need not commute in inhomogeneous settings. If a contextual hierarchy is erased before evolution or trajectory analysis, the resulting averaged model may preserve coarse mass while losing the handoff structure that organized the

local scene.

The cosmological fitting problem highlights the difficulty of relating local inhomogeneous structure to large-scale averaged descriptions. The nesting domain handoff problem is a broader diagnostic cousin. It asks how child scenes, parent contexts, and handoff supports should be declared, compared, and tested across gravitational scales.

Thus, the claim is not that existing gravitational tools are blind. The claim is that they do not usually extract parity-network hierarchy as a general *scene grammar*: a diagnostic layer that formalizes declared contributions, derived contexts, parent-child relations, confidence asymmetries, and modular handoff supports.

8 Relation to General Relativity

The proposed framework does not modify the Einstein field equations.

It does not introduce a new force.

It does not treat parity boundaries as physical surfaces.

It does not claim that spacetime is discrete.

It does not replace curvature, geodesics, or metric structure.

Instead, it proposes a diagnostic layer over declared gravitational scenes.

General relativity supplies the local geometric law. The present framework asks how source rosters, parent contexts, child domains, parity boundaries, and confidence asymmetries should be formalized when applying that law to multiscale systems.

This distinction is crucial.

A continuous field can still admit modular scene structure.

The modularity is not in the field as a discontinuity. It is in the declared relational organization of the scene.

The proposed framework therefore occupies a middle position: it is not new fundamental gravity, and it is not merely visualization.

It is a proposed mathematical grammar for gravitational context.

9 Tidal Eigenstructure as an Orientation Witness

Parity identifies scalar contribution-context balance. But a handoff is not purely scalar.

A child domain may become meaningful not only where its contribution equals the parent context, but where the local deformation frame begins to orient around the child.

For this we use the tidal tensor.

In the weak-field limit, define

$$\Phi(x) = -G \sum_i \frac{M_i}{r_i(x)}. \tag{19}$$

With metric approximation:

$$g_{00} \approx -(1 + 2\Phi/c^2), \tag{20}$$

and

$$g_{ij} \approx (1 - 2\Phi/c^2)\delta_{ij}. \tag{21}$$

The tidal tensor is

$$T_{ij}(x) = -\partial_i \partial_j \Phi(x). \tag{22}$$

For a point source outside the physical radius:

$$T_i(x) = \frac{GM_i}{r_i^3} \left(3\hat{r}_i \hat{r}_i^T - I\right). \tag{23}$$

The eigenvectors of T identify the principal local directions of stretch, compression, and shear. Let $e_0(T(x))$ be the dominant tidal eigenvector at x. For a child source c, let

$$\hat{r}_c(x) = \frac{x - x_c}{|x - x_c|}. \tag{24}$$

Define child alignment:

$$A_c(x) = |e_0(T(x)) \cdot \hat{r}_c(x)|. \tag{25}$$

Similarly, define parent alignment:

$$A_\Pi(x) = |e_0(T(x)) \cdot \hat{r}_\Pi(x)|. \tag{26}$$

An eigenvector handoff indicator is:

$$H_{\text{eig}}(x) = A_c(x) - A_\Pi(x). \tag{27}$$

When $H_{\text{eig}} > 0$, the dominant local tidal orientation is more child-aligned. When $H_{\text{eig}} < 0$, it is more parent-aligned.

This motivates the second major diagnostic claim:

> Tidal eigenvectors act as orientation witnesses for nesting-domain handoff.

They do not define handoff alone. They reveal whether local deformation is beginning to organize around the child domain rather than the enclosing parent context.

10 Handoff Functionals

The construction in this section is *scaffolding pending calibration.* It is included because the nesting-domain handoff problem requires more than a single scalar equality, but it should not be treated as a final invariant, canonical extraction rule, or field equation.

A credible handoff diagnostic should not depend on one scalar.

We propose a family of handoff functionals combining parity proximity, tidal dominance, eigenvector alignment, and parent-context uncertainty.

A first instance is:

$$H_{A|\Pi}(x) = w_1 \exp\left(-\frac{|\Psi_{A|\Pi}(x)|}{\Psi_{\text{scale}}}\right) + w_2 \log\left(\frac{\|T_A(x)\|}{\|T_\Pi(x)\| + \epsilon}\right) + w_3 \left(A_A(x) - A_\Pi(x)\right) - w_4 \Lambda_\Pi(x). \tag{28}$$

The first term rewards proximity to parity.

The second term measures child-parent tidal dominance.

The third term measures orientation handoff.

The fourth term penalizes unresolved parent context.

The weights w_k are not universal constants. They are diagnostic choices to be calibrated by stability, sensitivity, and predictive usefulness. Different applications may require different members of the handoff-functional family.

A candidate functional handoff support is then:

$$B_{A|\Pi} = \{x : H_{A|\Pi}(x) \text{ is locally maximal and stable under perturbation}\}. \tag{29}$$

The functional above should not be read as a universal invariant, a replacement for field equations, or a final law of handoff. It is a proposed diagnostic family. Its purpose is to make explicit the ingredients that a credible handoff claim must account for: scalar contribution balance, tensor organization, orientation transition, and uncertainty in the enclosing context. The weights, normalization scales, and stability criteria must be calibrated by controlled experiments. A handoff functional earns meaning only if it remains stable under perturbation and predicts or classifies independent diagnostics such as tidal eigenstructure, released-lattice deformation, or trajectory behavior.

The handoff is therefore not merely where one scalar equals another. It is where multiple diagnostic witnesses agree.

This is a new diagnostic relationship, not a new force law.

11 Gravitational Identity

A gravitational identity is not merely source membership. It is a stable relation between a declared contribution and its enclosing context.

We define the identity of a source or group A relative to parent context Π as the diagnostic tuple

$$I_A^{\Pi}(t) = \left[G_{A|\Pi}(t), H_{A|\Pi}(x,t), L_{A|\Pi}(t)\right], \tag{30}$$

where $G_{A|\Pi}$ is the parity handoff surface, $H_{A|\Pi}$ is the handoff field, and $L_{A|\Pi}$ is the confidence component of the relation. The multiplicative handoff confidence in Eq. (4) is therefore not a parallel construction; it is the proposed first form of the L component in Eq. (30).

In words:

> A gravitational identity is what remains coherent under declared context, resolution, and causal access.

This statement is philosophical, but not merely poetic. It follows from the formal structure. A child scene is not fully defined by its internal source roster alone. It is defined by its relation to a parent context, its parity handoff surface, its functional handoff support, and the confidence of that relation.

At this stage, Eq. (30) should be read as a labeled diagnostic tuple, not yet as a full algebra of identity. Composition under nesting, equivalence between scenes at different resolutions, and

refinement maps under finer source rosters are natural next formal steps. They are not required for the present paper's claim, and they are intentionally deferred.

This gives a new way to discuss objectivity in gravitational modeling.

The field quantities are not arbitrary. But gravitational identity is relationally stabilized.

12 Earth-Moon as a Minimal Example

The Earth-Moon system provides a useful minimal example because it contains a clear child-parent tension without requiring exotic assumptions.

Figure 2 gives a schematic version of the larger nesting logic in a toy Earth-Moon-Sun declaration. The drawing is not a simulation output. It illustrates the intended roles of the source-context parity boundary, Moon-local child refinement, and Earth-Moon group support relative to a parent solar context.

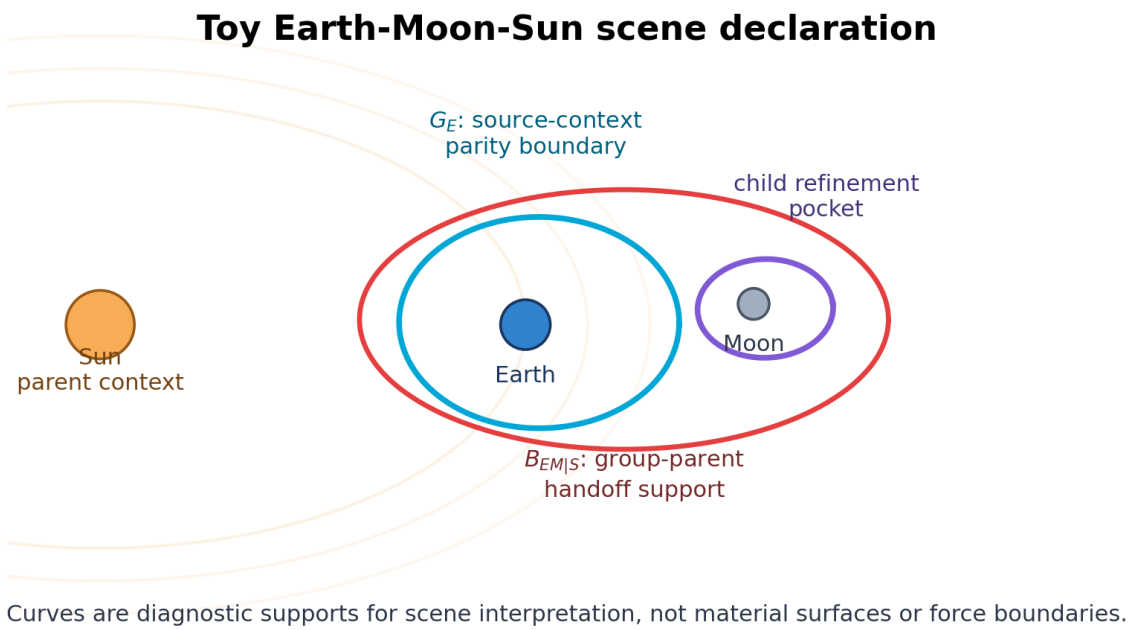


Figure 2: Toy Earth-Moon-Sun schematic showing how parity structures can organize a declared scene. The cyan curve indicates a source-context parity boundary for Earth, the purple curve indicates a Moon-local refinement pocket, and the red curve indicates a group-parent handoff support for the Earth-Moon child system relative to a solar parent context. The figure is conceptual, not a claim that these curves are physical membranes.

At the scale of a broad Earth-centered scene, Earth dominates the acceleration field over most of the Earth-Moon corridor. If one asks which source controls gross fall direction across the scene, Earth is the primary organizer. A global released lattice should therefore show Earth's dominant deformation. If the Moon appears as an equal global partner in such a view, the visualization is likely misleading.

Yet near the Moon, the Moon can dominate local tidal organization. This does not contradict

Earth's global dominance. It reveals the difference between acceleration dominance, tidal dominance, and child-domain identity.

A single declared scene may therefore contain both statements:

> Earth is the dominant parent organizer of the local scene.
>
> The Moon is a meaningful child organizer inside a local refinement domain.

This is the nesting problem in miniature.

If the Moon is treated only as a small perturbation of the Earth scene, its local tidal identity may be underrepresented. If the Moon is treated as isolated, the Earth-parent context is lost. A lawful scene declaration must preserve both the global parent deformation and the child refinement where Moon-local structure becomes meaningful.

A parity-network hierarchy provides a candidate scaffold for this relation. A group-parent parity boundary can identify where the Moon's declared contribution becomes readable against the Earth context. Tidal eigenstructure can test whether local deformation begins to organize around the Moon. A handoff functional can combine parity proximity, tidal dominance, eigenvector alignment, and parent-context confidence into a child-parent relation.

This example also clarifies why a single global grid is insufficient. A coarse global scene may correctly show Earth dominance while failing to resolve the Moon's local tidal organization. A nested refinement is not a cosmetic zoom. It is a lawful child scene whose boundary conditions should inherit from the parent context.

The key point is not that new forces are needed. The equations already provide the local field behavior. The missing structure lies in declaring, preserving, and testing the nested scene relationship.

13 The Three-Dominance Principle

The Earth-Moon example exposes a principle that should be kept explicit in any nested-scene analysis.

> A nested gravitational scene must distinguish acceleration dominance, tidal dominance, and identity dominance.

Acceleration dominance asks which source or group controls gross fall direction at a point or across a region. In weak-field notation this is primarily associated with $a(x) = -\nabla\Phi(x)$.

Tidal dominance asks which source or group controls local deformation structure. This is associated with the tidal tensor T_{ij} and its eigenstructure, not merely with the magnitude of acceleration.

Identity dominance asks which source or group remains relationally meaningful under the declared scene, parent context, parity hierarchy, and confidence record. This is not reducible to acceleration magnitude or tensor norm alone.

The three dominance notions may agree in simple settings, but nested scenes often become interesting precisely when they diverge. A child source can be globally subdominant in acceleration

while locally dominant in tidal organization and relationally meaningful as a child domain. This distinction is one reason parity-network hierarchy is useful: it gives a way to preserve source-context identity even when a single scalar field would prematurely flatten the scene.

14 Continuity-Preserving Released Lattices

A visual grid is not a physical object. It is a datum.

To visualize weak-field structure without implying that spacetime is a material substance, one may declare a coordinate lattice, sample the field at each lattice node, record the result, and render a controlled deformation.

A raw release displacement can be written:

$$u_{\text{raw}}(x) = \frac{1}{2} a(x) \tau^2, \tag{31}$$

where

$$a(x) = -\nabla \Phi(x), \tag{32}$$

and τ is a declared release parameter.

A rendered displacement may then be:

$$u_{\text{render}}(x) = A u_{\text{raw}}(x), \tag{33}$$

where A is a declared visualization gain.

However, raw release alone can produce spokes into source centers. A readable lattice requires neighborhood continuity. Therefore the visualization layer should distinguish raw release from continuity-preserving display.

A continuity-preserving released coordinate lattice can be defined as:

$$u_{\text{cont}} = \arg\min_u \int_\Omega \left[\|u(x) - u_{\text{raw}}(x)\|^2 + \lambda \|\nabla u(x)\|^2 \right] dx. \tag{34}$$

This expression preserves the field-derived release while penalizing discontinuous deformation.

For a fixed outer boundary:

$$u|_{\partial\Omega} = 0. \tag{35}$$

For a nested child refinement:

$$u_{\text{child}}|_{\partial\Omega_{\text{child}}} \approx \text{Interp}(u_{\text{parent}}). \tag{36}$$

The handoff mismatch can be measured as:

$$E_{\text{handoff}} = \int_{\partial\Omega_{\text{child}}} \|u_{\text{child}} - \text{Interp}(u_{\text{parent}})\|^2 \, dA. \tag{37}$$

This provides a diagnostic for whether a child refinement inherits the parent deformation smoothly.

The point is not that spacetime is a material lattice. The point is that a declared visualization datum can be made continuous, auditable, and downstream of field sampling.

15 Claims

This paper makes eight claims.

First, *scene declaration is mathematically meaningful.* A gravitational scene is not just a container for sources. It is a structured declaration of source roster, datum convention, readable field, parent context, causal access, and confidence.

Second, *parity networks expose modular structure in declared gravitational scenes.* This modularity is not a discontinuity in the field, but a relational decomposition of the scene.

Third, *nested child-parent domains require explicit handoff confidence.* A child scene and parent context are not generally known at the same resolution, freshness, or completeness.

Fourth, *parent-context uncertainty should be included in nested scene interpretation.* A parent may be necessary for interpretation while remaining unresolved at the child's fidelity.

Fifth, *tidal eigenvectors can serve as orientation witnesses for domain handoff.* They help determine whether local deformation remains parent-organized or begins to organize around a child domain.

Sixth, *nested scenes require a distinction between acceleration dominance, tidal dominance, and identity dominance.* These are related but not interchangeable quantities.

Seventh, *handoff functionals can combine multiple diagnostic witnesses.* Parity, tidal dominance, eigenvector alignment, and parent-context uncertainty can be integrated into candidate diagnostics for multiscale gravitational nesting, pending calibration.

Eighth, *continuity-preserving released coordinate lattices provide a disciplined visualization method.* They allow weak-field metric deformation to be visualized without confusing the visualization datum with a physical substance.

These are new diagnostic and mathematical relationships. They do not require new force laws.

16 Predictions

The proposed framework makes five testable predictions.

First, parity-network hierarchies should remain stable across controlled perturbations in well-posed weak-field scenes. Perturbations include ray count, mesh density, source position, softening or cutoff rules, parent roster, and datum convention.

Second, parity boundaries should correlate with transitions in tidal eigenstructure. A parity boundary is not a physical wall, but it may mark regions where scalar source-context balance, tidal dominance, and eigenvector orientation begin to reorganize.

Third, nested child domains should require local refinement even when globally subdominant. A child source may contribute weakly to global acceleration while dominating local tidal organization in its refinement pocket.

Fourth, premature averaging should erase handoff structure. Treating a multiscale source group as a single bulk source may preserve total mass while destroying parity hierarchy, tidal orientation

transitions, and contextual identity.

Fifth, trajectory diagnostics should respond to retained parity hierarchy. If parity networks capture real contextual structure, then trajectory segments, geodesic bundles, or weak-field test paths should show systematic changes across retained handoff regions.

These predictions define the immediate research program. The framework succeeds only if these predictions survive sensitivity testing.

The first targeted validation should focus on H_{eig}. In the weak-field setting of this paper, H_{eig} is computed from the Newtonian tidal tensor T_{ij}. The natural relativistic upgrade is to compute the same dominant-eigenvector alignment using the electric part of the Weyl tensor, E_{ij}, in a regime where the two descriptions should agree [7]. A controlled Earth-Moon-Sun comparison of $H_{\mathrm{eig}}[T]$ and $H_{\mathrm{eig}}[E]$ would test whether the orientation-witness diagnostic survives the upgrade from weak-field tidal structure to the corresponding GR curvature object. A null difference would be valuable: it would license the use of T_{ij} as a calibrated weak-field proxy rather than a convenience.

17 What Is Not Being Claimed

This paper does not claim that parity boundaries are physical membranes.

It does not claim that parity networks are observables.

It does not claim a new force.

It does not claim a new particle.

It does not claim that spacetime is literally a lattice, sponge, or material medium.

It does not claim a replacement for the Einstein field equations.

It does not claim that general relativity is wrong in its validated domain.

The claim is more precise:

> Declared gravitational scenes contain modular relational structure that standard workflows do not usually extract as a first-class diagnostic.

That structure may matter.

18 Discussion: Modularity Without Discontinuity

A modular gravitational scene may sound paradoxical. General relativity is continuous. The field does not come pre-cut into human-friendly pieces.

But modularity here does not mean discontinuity.

It means that a declared scene admits derived relational structure. A continuous gravitational field can still contain source-context relationships, child domains, parent supports, and handoff regions. These are not imposed arbitrarily. They are extracted from declared contribution fields, parity relations, tidal organization, confidence records, and stability tests.

The proposed framework therefore separates two ideas that are often conflated: the continuity of the field and the modularity of scene interpretation.

Parity networks do not fracture the field. They reveal how the declared scene organizes itself relationally.

This matters because many multiscale systems may lose essential structure under premature averaging. If a parent-child handoff is averaged away, the local equations may still be correct while the modeled context becomes too coarse to preserve the structure that mattered.

The problem is not the local law.

The problem is the bookkeeping of context.

19 Conclusion

The nesting domain handoff problem reframes a familiar but underformalized difficulty in multiscale gravitational modeling.

The challenge is not only to compute a field. It is to declare what scene the field belongs to, what parent context supports it, what child domains emerge inside it, what structure survives averaging, and where interpretation changes scale.

General relativity supplies the local law. This framework proposes a diagnostic grammar over declared scenes: a parity-network hierarchy capable of identifying modular gravitational context, child-parent handoff supports, confidence asymmetries, and regions where local tidal organization changes.

The local equations may be correct while inherited bookkeeping of gravitational context remains incomplete.

This is the central claim.

A continuous gravitational field can still admit modular scene structure. A child domain can be locally meaningful while remaining embedded in a larger parent context. A parity boundary can be mathematically useful without being a physical membrane. A handoff relation can be real as a diagnostic without being a new force.

If parity-network hierarchy proves stable and predictive across controlled scenes, it may supply a conservative but powerful addition to GR-compatible modeling: not new gravity, but a clearer mathematics of gravitational context.

The next step is empirical discipline. The framework must be tested against tidal eigenstructure, released-lattice deformation, trajectory diagnostics, sensitivity sweeps, and averaging comparisons. If it fails, it fails as a diagnostic proposal. If it holds, then multiscale gravitational modeling gains a missing grammar: a way to say not only what the field is, but what scene the field belongs to.

A Relation to Papers I-IV

This paper depends on, but also reframes, the preceding four-paper sequence.

Paper I introduced the gravitational coherence surface as a parity horizon of gravitational distinguishability under a declared readability rule. In the present paper, that object becomes the special case $G_i = \{x : \Psi_i(x) = 0\}$: the parity boundary where a declared contribution reaches equality with its derived context.

Paper II generalized the construction into many-source shared parity networks. In the present paper, those networks are interpreted as modular scene-decomposition diagnostics. Pairwise and

groupwise surfaces are no longer treated only as extracted boundary geometry; they become candidate supports in a hierarchy of source-context relationships.

Paper III argued that mesoscale structure can persist between pointwise field evaluation and coarse averaging. In the present paper, that mesoscale persistence is connected to parent-child nesting: premature averaging may erase parity hierarchy, tidal-orientation transition, and handoff structure.

Paper IV reported proof-of-concept extraction of many-source weak-field boundary-support geometry. In the present paper, those extraction results motivate the broader question of whether extracted parity hierarchy predicts or classifies independent diagnostics such as tidal eigenstructure, released-lattice deformation, and trajectory behavior.

Thus, Paper V is not a replacement for the GCS sequence. It is the first attempt to interpret the sequence as a context calculus: a formal grammar for declared scenes, parent contexts, child domains, handoff confidence, and modular gravitational identity.

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